HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed bare-metal firmware in **Embedded C++** for a new medical device, focusing on low-level hardware interaction and board bring-up.
- Implemented **Bluetooth Low Energy (BLE) connectivity** for device-to-mobile communication, ensuring reliable data transmission and secure pairing.
- Integrated embedded firmware with hardware components, performing detailed **board debugging** and signal analysis to resolve integration issues.
- Collaborated with a cross-functional team including mobile engineers and testers to define and implement firmware requirements for the PureWick device programme.
- Participated in **code reviews** for embedded C/C++ modules, ensuring adherence to coding standards and best practices within the team.
- Contributed to technical estimations and sprint planning sessions, providing input on the complexity and effort for firmware development tasks.
- Containerized firmware build and testing tools using **Docker** to create reproducible development environments and streamline CI/CD integration.

SKILLS

Languages – Embedded C, Embedded C++

Connectivity – Bluetooth Low Energy (BLE)

Development Tools – Bare-metal firmware development, Board bring-up, Debugging, Integration, Docker

Methodologies – Agile, Scrum, Code Reviews

Collaboration – Cross-functional teams, Mobile engineering, Testing, DevOps

PROJECTS

PureWick Device Firmware Development (HCL Technologies)

- Engineered **bare-metal firmware** in Embedded C++ for a next-generation medical device, focusing on direct hardware control and system initialization.
- Designed and implemented the **Bluetooth Low Energy (BLE) stack** to manage device discovery, connection, and secure data exchange with companion mobile applications.
- Conducted thorough **board bring-up and debugging** using JTAG and logic analyzers, resolving critical hardware-firmware integration challenges to achieve stable operation.

Embedded Systems Integration & Testing (HCL Technologies)

- Developed Python scripts to automate firmware flashing and basic functional tests across multiple device prototypes, reducing manual testing time by **30%**.
- Integrated firmware modules with sensor inputs and communication interfaces, performing system-level validation against hardware specifications.
- Documented integration procedures and test results, contributing to the knowledge base for future embedded development cycles.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)